

03 Matrix Multiplication

3.2 The Second View of Matrix Multiplication

Outer-Product Expansion, Sums of Rank-One Matrices, and Heatmap Visualization



Linear Algebra Made Easy and Visual | Open-source code & drafts: <https://github.com/Visualize-ML>

Core Skills for This Section

1

The outer-product expansion view: matrix multiplication can be written as a sum of several outer-product matrices

2

Every outer product is a rank-one matrix

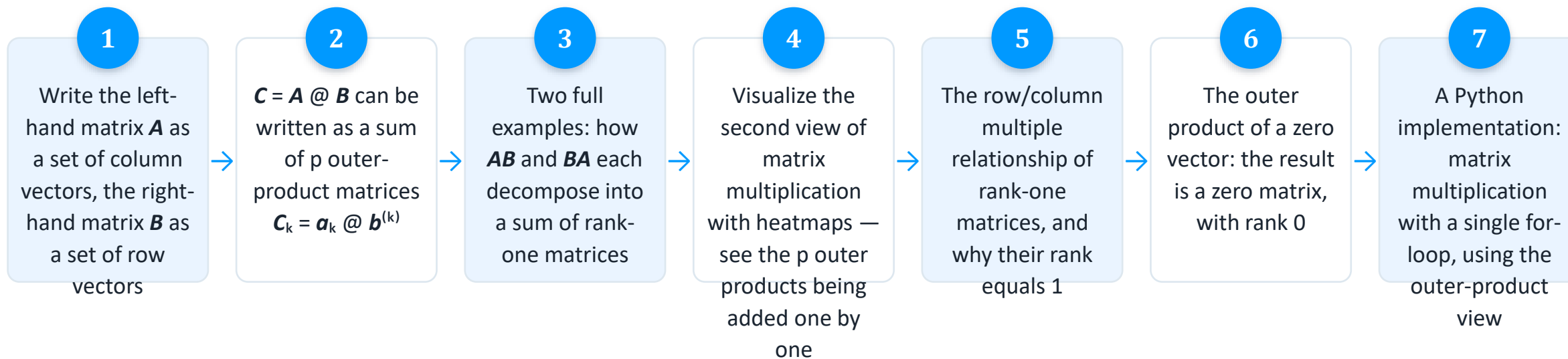
3

Rank-one matrices have row/column multiple relationships: this reflects linear dependence

4

The outer product of a zero vector gives a zero matrix

How This Section Unfolds



Breaking Matrix Multiplication into a Sum of Matrices

The second view of matrix multiplication writes the left-hand matrix as column vectors and the right-hand matrix as row vectors — it is, in essence, a sum of outer products of column and row vectors

- The previous section showed that matrix multiplication can be “expanded by inner products”; this section shows it can equally be “expanded by outer products”
- This decomposition lets us understand the structure and meaning of matrix multiplication at a deeper level

Writing Matrix $\mathbf{A}$ as a Set of Column Vectors

$$\mathbf{A}_{m \times p} = [\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_p]$$

- For matrix multiplication $\mathbf{A} @ \mathbf{B}$, write the left-hand matrix $\mathbf{A}$ as a set of column vectors
- Column vector $\mathbf{a}_k$ ($k = 1, 2, \dots, p$) has shape $m \times 1$

Figure 1 · Writing the Left-Hand Matrix A as a Set of Column Vectors

- A is an $m \times p$ matrix, split into p column vectors $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_p$, each of shape $m \times 1$
- These p column vectors will each be paired with a corresponding row vector of B to form p outer products

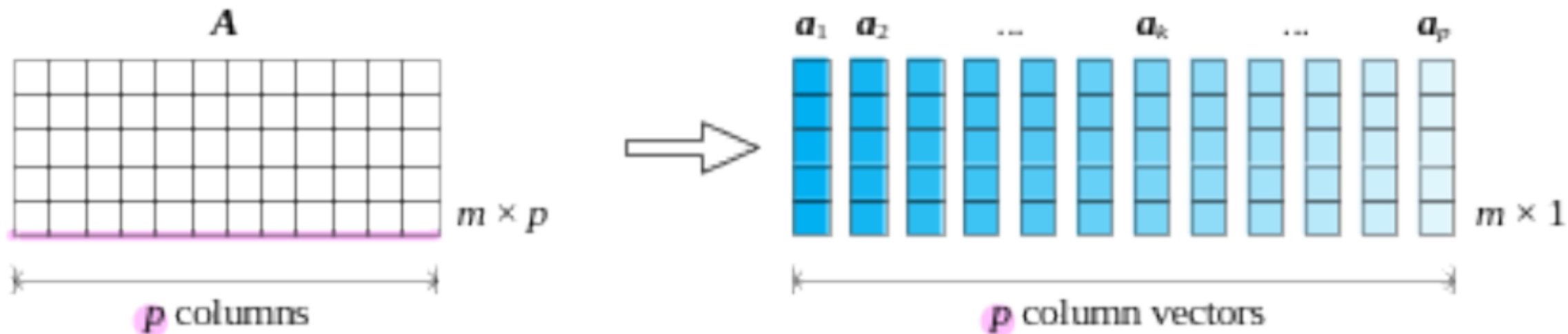


图 1. 把左侧矩阵 A 写成一组列向量

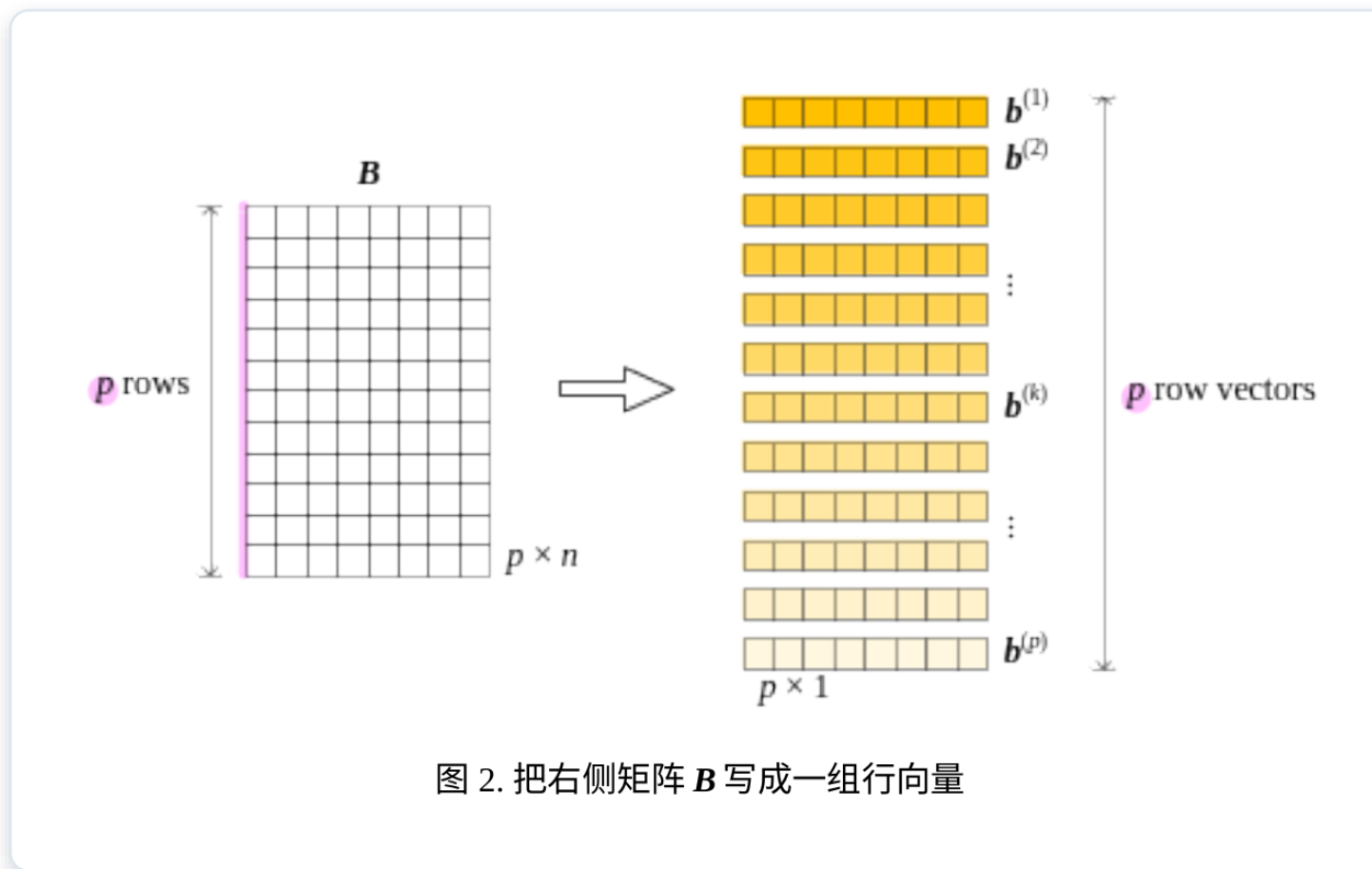
Writing Matrix $\mathbf{B}$ as a Set of Row Vectors

$$\mathbf{B}_{p \times n} = [\mathbf{b}^{(1)}; \mathbf{b}^{(2)}; \vdots; \mathbf{b}^{(p)}]$$

- Write the right-hand matrix $\mathbf{B}$ as a set of row vectors; row vector $\mathbf{b}^{(k)}$ has shape $1 \times n$
- $\mathbf{A}$'s column count p and $\mathbf{B}$'s row count p correspond exactly — this is the dimension that gets “canceled out” in matrix multiplication

Figure 2 · Writing the Right-Hand Matrix B as a Set of Row Vectors

- B is a $p \times n$ matrix, split into p row vectors $\mathbf{b}^{(1)}, \mathbf{b}^{(2)}, \dots, \mathbf{b}^{(p)}$, each of shape $1 \times n$
- Next, pairing $\mathbf{a}_k$ with $\mathbf{b}^{(k)}$ one by one lets us write $\mathbf{A} @ \mathbf{B}$ as a sum of outer products



A @ B Can Be Written as a Sum of Matrices with the Same Shape

$$\mathbf{C} = \mathbf{A} @ \mathbf{B} = \mathbf{a}_1 @ \mathbf{b}^{(1)} + \mathbf{a}_2 @ \mathbf{b}^{(2)} + \dots + \mathbf{a}_p @ \mathbf{b}^{(p)} = \sum_k \mathbf{a}_k @ \mathbf{b}^{(k)}$$

- Let $\mathbf{C}_k = \mathbf{a}_k @ \mathbf{b}^{(k)}$ — this is the outer product introduced in the previous section; $\mathbf{C}$ and $\mathbf{C}_k$ share exactly the same shape, $m \times n$
- Matrix multiplication $\mathbf{A} @ \mathbf{B}$ turns into a sum of p matrices — this is the second view of the matrix multiplication rule, outer-product expansion

Figure 3 · $A @ B$ Can Be Written as a Sum of Matrices with the Same Shape

- $a_k @ b^{(k)} = C_k$: $(m \times 1) @ (1 \times n)$ has the shared dimension 1 “canceled out,” leaving a result of shape $m \times n$
- The p matrices C_k , all of the same shape, are added one by one to finally produce $C = \sum_k C_k$

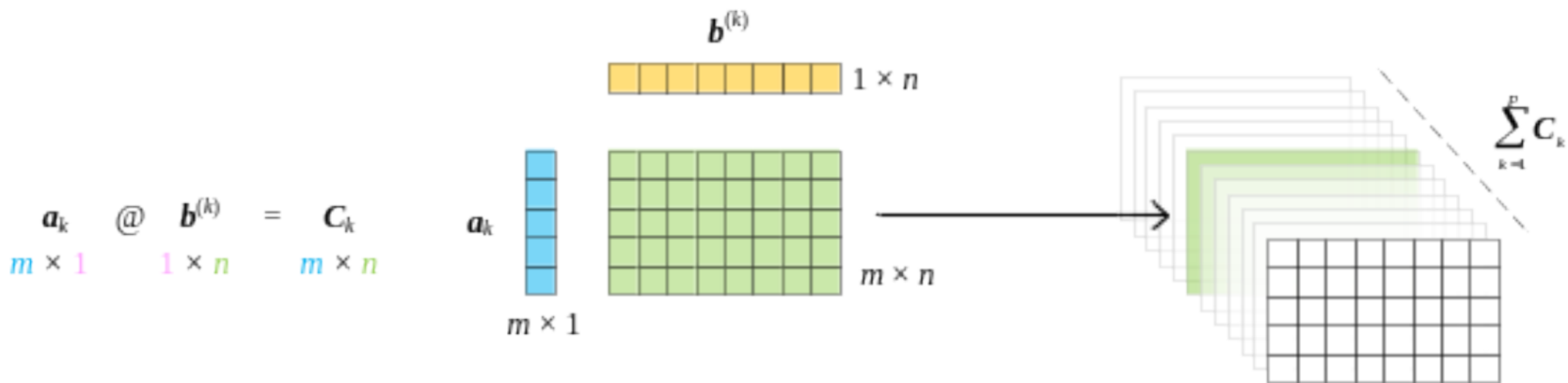


图 3. $A @ B$ 可以写成一组相同形状矩阵的求和

Back to the Earlier Example: $A @ B$

$A = [1,2,3; 4,5,6]$, $B = [1,4; 2,5; 3,6]$; with the left written as column vectors and the right as row vectors, $A @ B = \mathbf{a}_1 @ \mathbf{b}^{(1)} + \mathbf{a}_2 @ \mathbf{b}^{(2)} + \mathbf{a}_3 @ \mathbf{b}^{(3)}$

- $\mathbf{a}_1 = [1,4]^T$, $\mathbf{a}_2 = [2,5]^T$, $\mathbf{a}_3 = [3,6]^T$; $\mathbf{b}^{(1)} = [1,4]$, $\mathbf{b}^{(2)} = [2,5]$, $\mathbf{b}^{(3)} = [3,6]$
- By shape: $(2 \times 1) @ (1 \times 2)$ has the shared dimension 1 “canceled out,” so each outer-product result has shape 2×2

Figure 4 · Computing C_1 , C_2 , and C_3 Separately

- $C_1 = \mathbf{a}_1 @ \mathbf{b}^{(1)} = [1, 4; 4, 16]$, $C_2 = \mathbf{a}_2 @ \mathbf{b}^{(2)} = [4, 10; 10, 25]$, $C_3 = \mathbf{a}_3 @ \mathbf{b}^{(3)} = [9, 18; 18, 36]$
- Every C_k is a rank-one matrix: its rows are multiples of one another, and so are its columns

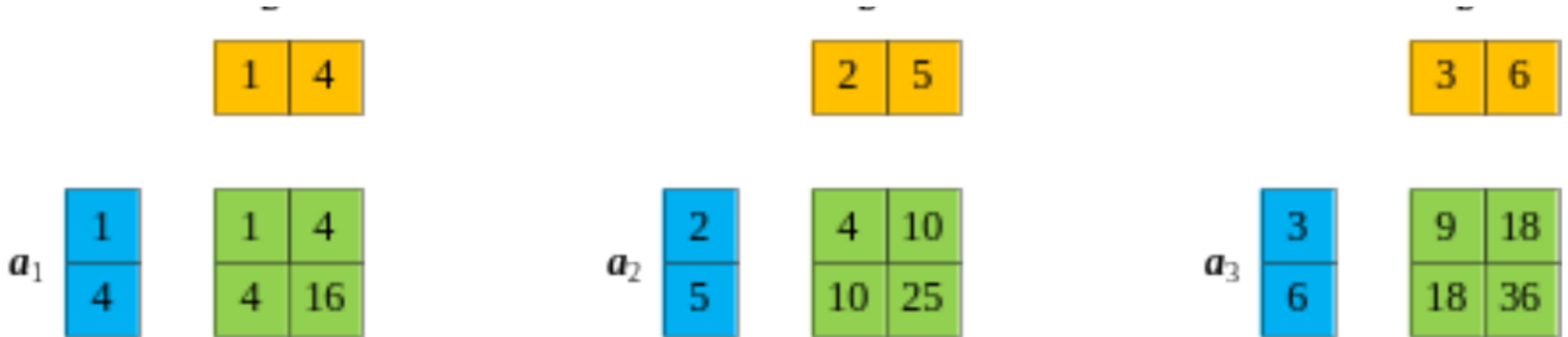


图 4. 分别计算 C_1 、 C_2 、 C_3

Figure 5 · Superposing C1, C2, and C3

- $C = C_1 + C_2 + C_3 = [1,4;4,16] + [4,10;10,25] + [9,18;18,36] = [14,32;32,77]$
- This matches the result from the first view (inner-product expansion) in the previous section exactly — different paths, same destination

$$C_1 = \mathbf{a}_1 @ \mathbf{b}^{(1)}$$

1	4
4	16

+

$$C_2 = \mathbf{a}_2 @ \mathbf{b}^{(2)}$$

4	10
10	25

+

$$C_3 = \mathbf{a}_3 @ \mathbf{b}^{(3)}$$

9	18
18	36

=

 C

图 5. C_1 、 C_2 、 C_3 三个矩阵叠加

Now the Other Order: $B @ A$

With the left-hand matrix B written as column vectors and the right-hand matrix A as row vectors, $B @ A = b_1 @ a^{(1)} + b_2 @ a^{(2)}$

- $b_1 = [1, 2, 3]^T$, $b_2 = [4, 5, 6]^T$; $a^{(1)} = [1, 2, 3]$, $a^{(2)} = [4, 5, 6]$
- By shape: $(3 \times 1) @ (1 \times 3)$ has the shared dimension 1 “canceled out,” so each outer-product result has shape 3×3

Figure 6 · Computing D1 and D2 Separately

- $D_1 = b_1 @ a^{(1)} = [1,2,3;2,4,6;3,6,9]$, $D_2 = b_2 @ a^{(2)} = [16,20,24;20,25,30;24,30,36]$
- A has only 2 rows and B only 2 columns, so $B @ A$ needs just two outer products to superpose, while $A @ B$ needed three

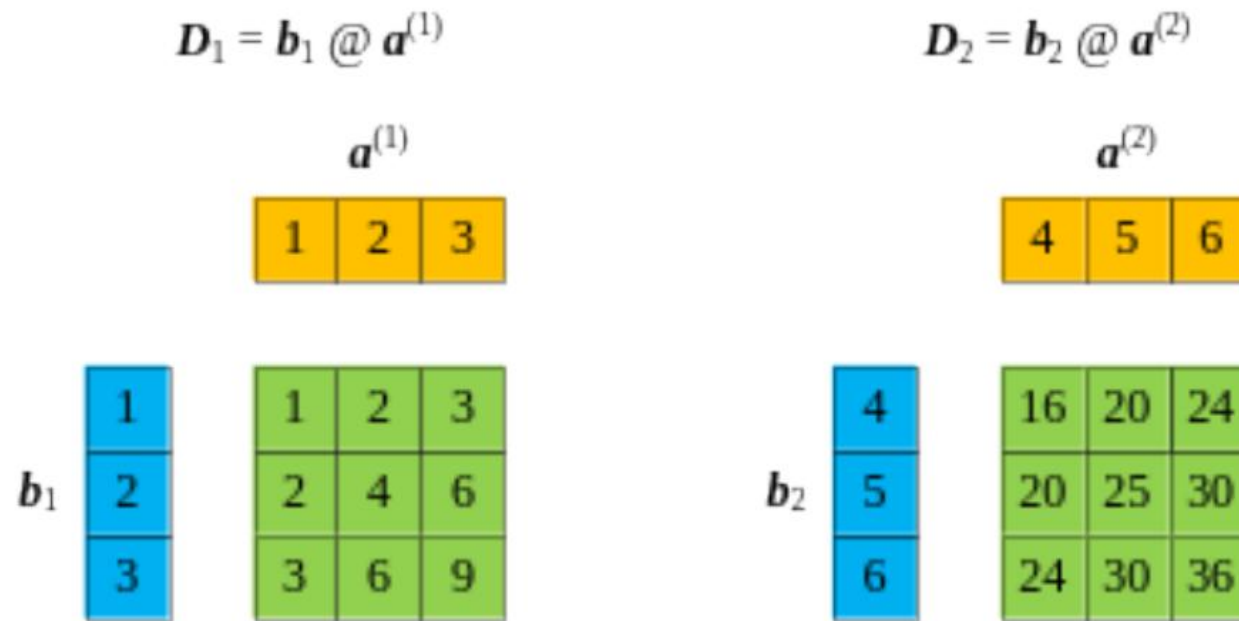


图 6. 分别计算 D_1 、 D_2

Figure 7 · Superposing D1 and D2

- $D = D_1 + D_2 = [17, 22, 27; 22, 29, 36; 27, 36, 45]$, matching the result from the first view exactly
- The number of outer products needed for the superposition, p , is exactly A 's column count (B 's row count) — the dimension that gets “canceled out”

$$D_1 = \mathbf{b}_1 @ \mathbf{a}^{(1)} \quad D_2 = \mathbf{b}_2 @ \mathbf{a}^{(2)}$$

1	2	3
2	4	6
3	6	9

+

16	20	24
20	25	30
24	30	36

=

17	22	27
22	29	36
27	36	45

图 7. D_1 、 D_2 两个矩阵叠加

Using Heatmaps to Show the Second View of Matrix Multiplication

Color intensity represents the magnitude of each entry; heatmaps let us “watch” p outer products being added one by one to assemble $A @ B$

- Given a pair of matrices A, B with 12 matching columns/rows, first look at the overall heatmap of $A @ B$ under the first view
- Then display the heatmap of every outer product $a_k @ b^{(k)}$ one by one, to get an intuitive feel for the “superposition” process

Figure 8 · Heatmap of Matrix Multiplication $A @ B$, the First View

- A , B , and $A @ B$ are all shown as heatmaps, with color corresponding to entry value
- This is the overall picture under the first view of matrix multiplication; next we break it apart into a series of outer-product heatmaps

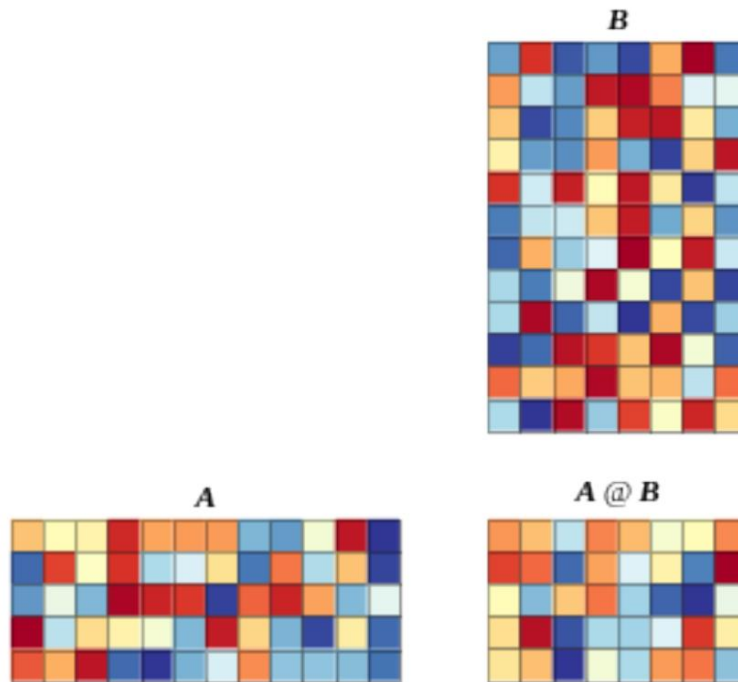


图 8. 矩阵乘法 $A @ B$ 热图，第一视角

Figure 9 · Heatmap Showing $\mathbf{a}_1 @ \mathbf{b}^{(1)}$

- $\mathbf{a}_1$ is $\mathbf{A}$'s first column (a color gradient from light for negative to dark for positive), and $\mathbf{b}^{(1)}$ is $\mathbf{B}$'s first row
- Their outer product $\mathbf{a}_1 @ \mathbf{b}^{(1)}$ produces a rank-one matrix heatmap with the same shape as $\mathbf{A} @ \mathbf{B}$

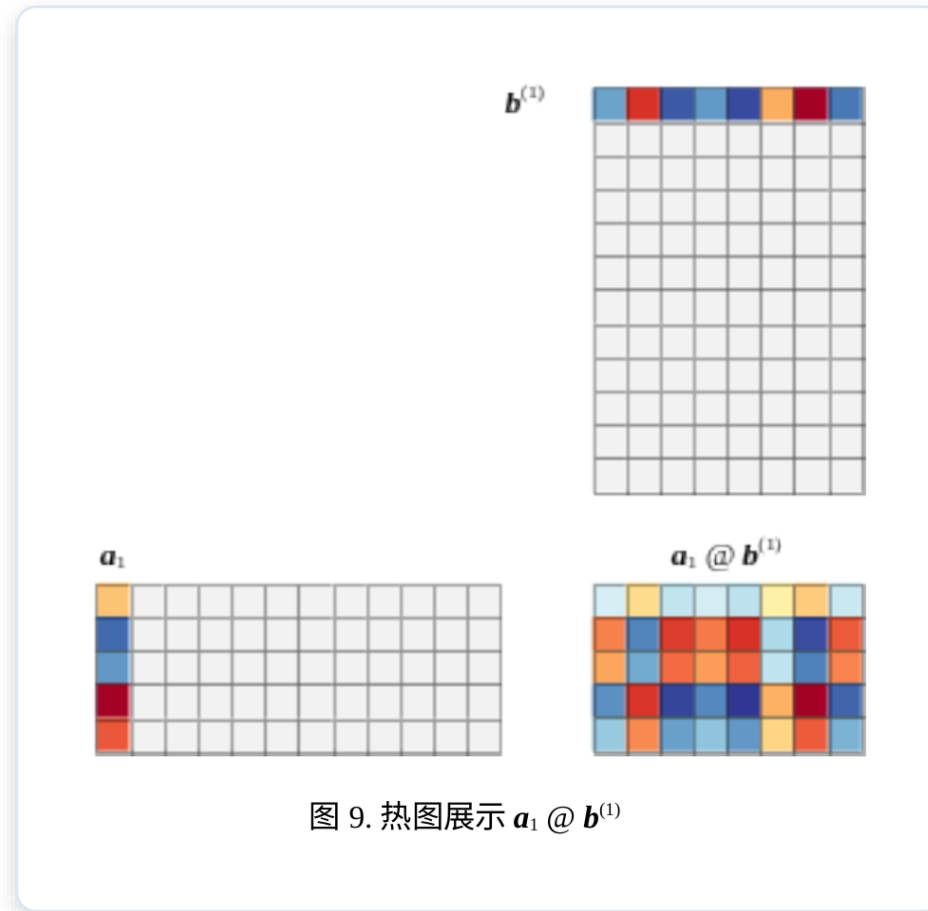


Figure 10 · The 12 Outer Products $\mathbf{a}_k @ \mathbf{b}^{(k)}$

- $\mathbf{A}$ has 12 columns ($\mathbf{B}$ has 12 rows), so there are 12 outer products altogether, just like $\mathbf{a}_1 @ \mathbf{b}^{(1)}$
- Adding these 12 rank-one matrices one by one produces exactly the $\mathbf{A} @ \mathbf{B}$ shown in Figure 8

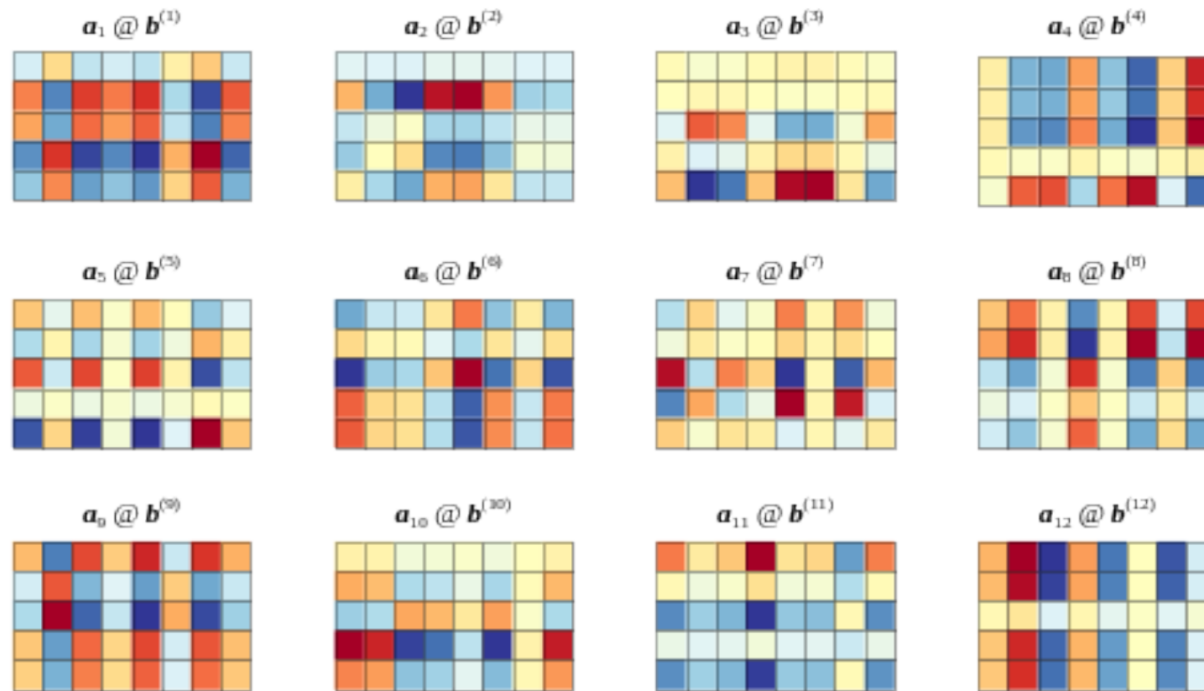


图 10. 12 个 $\mathbf{a}_k @ \mathbf{b}^{(k)}$

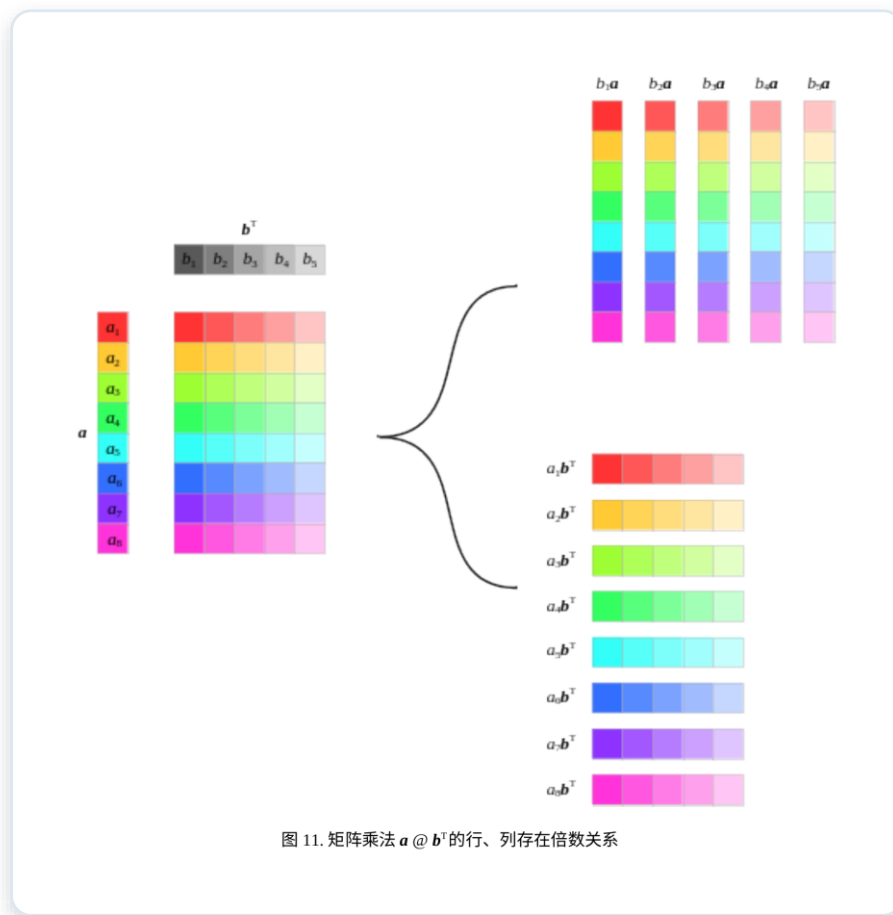
The Rows and Columns of the Outer Product $\mathbf{a} @ \mathbf{b}^T$ Are All Multiples

Given two column vectors $\mathbf{a}$, $\mathbf{b}$, the rows and columns of matrix multiplication $\mathbf{a} @ \mathbf{b}^T$ both exhibit multiple relationships

- Column k of $\mathbf{a} @ \mathbf{b}^T$ equals b_k times the whole vector $\mathbf{a}$: $\mathbf{a} @ \mathbf{b}^T = [b_1\mathbf{a}, b_2\mathbf{a}, \dots, b_n\mathbf{a}]$
- Row k of $\mathbf{a} @ \mathbf{b}^T$ equals a_k times the whole row vector $\mathbf{b}^T$: every row is a multiple of $\mathbf{b}^T$

Figure 11 · The Rows and Columns of Matrix Multiplication $\mathbf{a} @ \mathbf{b}^T$ Are Multiples

- Top right: every column of $\mathbf{a} @ \mathbf{b}^T$ is a multiple of $\mathbf{a}$ ($b_1\mathbf{a}$, $b_2\mathbf{a}$, ...); bottom right: every row is a multiple of $\mathbf{b}^T$ ($a_1\mathbf{b}^T$, $a_2\mathbf{b}^T$, ...)
- It's precisely because of this row/column multiple relationship that we call an outer product like $\mathbf{a} @ \mathbf{b}^T$ a rank-one matrix



Rank-One Matrices: Linear Dependence Between Rows and Columns

The rank of a matrix is the maximum number of linearly independent vectors among its rows or columns; a rank-one matrix has rank 1

- Every row of a rank-one matrix is a multiple of the same row vector; every column is a multiple of the same column vector — this topic will be covered in far more depth later in the book
- Note: if $\mathbf{a}$ or $\mathbf{b}$ is the zero vector $\mathbf{0}$, then $\mathbf{a} @ \mathbf{b}^T$ results in the zero matrix $\mathbf{O}$, and the matrix's rank is 0

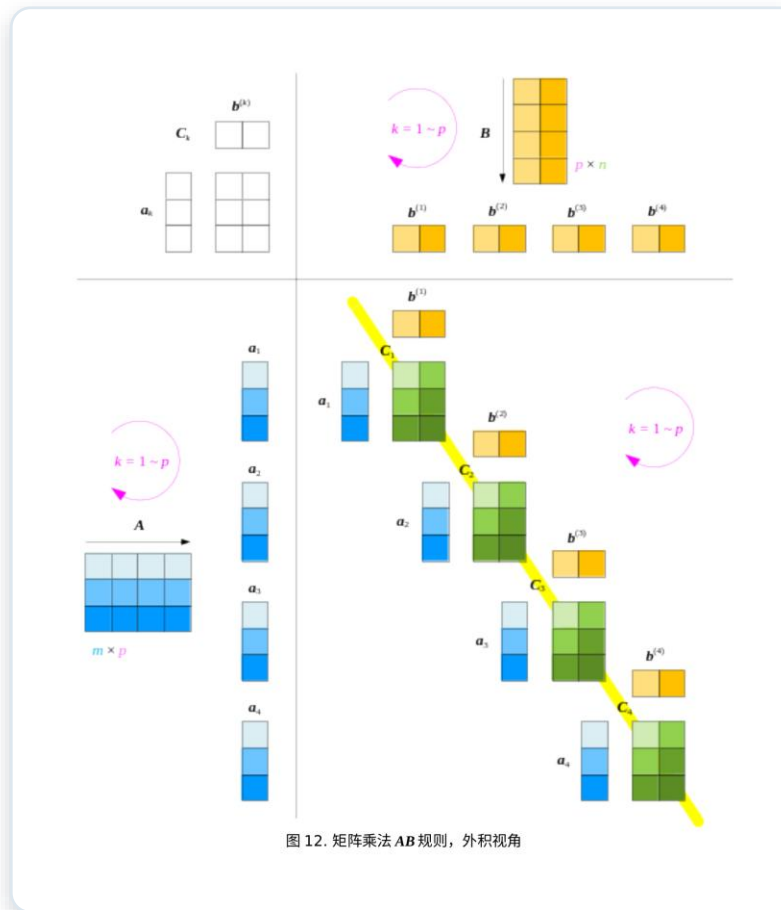
The Outer-Product View: One For-Loop Accumulating p Rank-One Matrices

For $k = 1$ to p , take A 's k -th column a_k and B 's k -th row $b(k)$, compute the outer product $a_k @ b(k)$, and accumulate it into result matrix C

- Unlike the nested loop of the inner-product view from the previous section, the outer-product view needs only a single loop — each pass produces a complete $m \times n$ matrix that's added directly to the accumulator
- This “accumulate rank-one matrices” idea will reappear again and again in later topics such as matrix decomposition and low-rank approximation

Figure 12 · The Rule for AB, the Outer-Product View

- The loop variable k runs from 1 to p ; each step takes A 's k -th column a_k and B 's k -th row $b^{(k)}$, and computes outer product C_k
- As shown along the yellow diagonal, $C_1, C_2, \dots, C_p$ are accumulated one by one to finally assemble the complete $C = A @ B$



Code 1 · A Custom Python Function for Matrix Multiplication

LA_03_02_03.ipynb

- `def matrix_multiplication_outer(A, B)` first reads the shapes, checks whether `p_A` equals `p_B`, and raises an error if they don't match
- It initializes result matrix `C` as an (m, n) matrix of zeros, then runs a single for-loop from $k = 0$ to $p_A - 1$
- `a_k = A[:, [k]]` takes `A`'s k -th column (a column vector), `b_k = B[[k], :]` takes `B`'s k -th row (a row vector), and `C += (a_k @ b_k)` accumulates this step's outer product into the result

```
## 初始化
import numpy as np

## 自定义函数，矩阵乘法第二视角
def matrix_multiplication_outer(A, B):

    # 获取矩阵 A 和 B 的形状
    m, p_A = A.shape
    p_B, n = B.shape

    # 检测矩阵形状是否符合矩阵乘法规则
    if p_A != p_B:
        raise ValueError('Dimensions do not match')

    # 初始化结果矩阵 C，形状 (m, n)，初始值设为 0
    C = np.zeros((m, n))

    for k in range(p_A):
        a_k = A[:, [k]]      # A 的第 k 列，二维数组
        b_k = B[[k], :]      # B 的第 k 行，二维数组
        C += (a_k @ b_k)     # 每次计算外积并加到结果中

    return C

## 矩阵乘法
A = np.array([[1, 2, 3],
              [4, 5, 6]])
B = A.T

## 矩阵乘法
matrix_multiplication_outer(A, B)
matrix_multiplication_outer(B, A)
```

Outer-Product Expansion: The Second Key to Matrix Multiplication

Split A into column vectors, split B into row vectors, pair them up one by one to form outer products, then add them together — this is the second key to understanding the structure of matrix multiplication

- Every outer product is a rank-one matrix with row/column multiple relationships — laying the groundwork for later topics on rank and linear dependence
- The next section introduces the third view of matrix multiplication, enriching our understanding of matrix multiplication still further

Key Terms in This Section

Outer-Product Expansion $A @ B$ can be written as a sum of p outer-product matrices $a_k @ b(k)$

Rank The maximum number of linearly independent vectors among a matrix's rows or columns

Linear Independence No non-trivial linear combination of the vectors equals the zero vector

Zero-Vector Outer Product If a or b is the zero vector, $a @ b^T$ results in a zero matrix with rank 0

Heatmap A visualization using color intensity to represent the magnitude of matrix entries

Rank-One Matrix $a @ b^T$ — a matrix whose rows and columns are all multiples of one another

Linear Dependence A multiple or linear-combination relationship exists between vectors

Superposition Writing C as a sum of p same-shape matrices C_k

The Canceled Dimension p A 's column count, B 's row count — the number of terms summed in outer-product expansion

Single For-Loop (Outer-Product View) Iterating $k = 1$ to p , computing and accumulating one rank-one matrix per step

GO FURTHER

Companion Code and Further Reading

Further Reading · LA_03_02_01.ipynb

Computes both worked examples in this section using the second view of matrix multiplication

Further Reading · LA_03_02_02.ipynb

Draws all the heatmaps in this section, visualizing the outer-product expansion process

Further Reading · LA_03_02_03.ipynb

A custom Python function for matrix multiplication, implemented from the outer-product view

Open-Source Repository

Code and figures for the whole book: <https://github.com/Visualize-ML>

Questions Q1 – Q9

Q1 Given the following four pairs of matrices A and B, use the second view of matrix multiplication to compute $A @ B$ and $B @ A$ for each pair:

① $A=[1,2;3,4]$, $B=[1,3;2,4]$; ② $A=[1,0;3,1]$, $B=[1,0;2,1]$; ③ $A=[1,1;0,1]$, $B=[1,3;0,1]$; ④ $A=[1,2;3,4;5,6]$, $B=A^T$.

Q3 What is the rank of a matrix, and is it related to the matrix's size?

Q5 What is linear independence? How can it be understood geometrically?

Q7 Following Figures 8-10, draw heatmaps for a randomly generated pair of 4×3 and 3×5 matrices, using matplotlib's `imshow()` to show each outer product and their cumulative sum.

Q9 Write a function `is_rank_one(M)` that determines whether a given matrix M can be written as the outer product of two vectors (i.e., verify the row/column multiple relationship).

Q2 Given the following pairs of column vectors, write the matrix multiplication $a @ b^T$ as row/column multiple relationships: ① $a=[1,2]^T$, $b=[1,2]^T$; ② $a=[1,2]^T$, $b=[1,2,3]^T$; ③ $a=[1,2,3]^T$, $b=[1,2]^T$.

Q4 What is linear dependence? How can it be understood geometrically?

Q6 Write a function `outer_product_sum(A, B)` that computes matrix multiplication using outer-product expansion (the second view), and compare it against $A @ B$ on several randomly shaped matrices.

Q8 For this section's first example, write a loop that prints $C1$, $C1+C2$, and $C1+C2+C3$ in turn, to verify how the result gradually approaches the final product.

Questions Q10 – Q18

Q10 Use `np.linalg.matrix_rank()` to verify that C1, C2, and C3 from the first example are all rank-one matrices.

Q12 Time `matrix_multiplication_inner` (the previous section's inner-product view) against `matrix_multiplication_outer` (this section's outer-product view) on large matrices, and compare their runtimes.

Q14 For the second example ($B @ A$), use Python to verify that D exactly equals $D1 + D2$.

Q16 Write a function `estimate_rank_from_svd(M)` that uses `np.linalg.svd()` to count the number of significant singular values and estimate matrix M's rank.

Q18 Construct three different (a, b) pairs where a or b is the zero vector, and use Python to verify that $a @ b^T$ always results in an all-zero matrix.

Q11 Construct an example where one column vector a_k is the zero vector, and verify whether the corresponding C_k is a zero matrix.

Q13 Write a function `decompose_and_verify(A, B)` that decomposes $A @ B$ into p rank-one matrices, sums them back up, and uses `np.allclose()` to check the result matches $A @ B$.

Q15 Use Python to verify the row/column multiple relationships in Figure 11: for chosen a and b, print out each column and row of $a @ b^T$, confirming every column is a multiple of a and every row a multiple of b^T .

Q17 Modify Code 1 to print the intermediate result C after each accumulation step inside the loop, and observe how the matrix gets “filled in” step by step.

Questions Q19 – Q25

Q19 For the four matrix pairs in Q1, write a script that automatically computes and prints results from the inner-product view (previous section) and the outer-product view (this section), verifying the two always agree.

Q21 Explore whether the sum of two rank-one matrices is still rank-one: test with several random examples and explain what you find.

Q23 Write a function that compares $\|A @ B\|_F$ against the sum of the Frobenius norms of its individual rank-one components, and explain whether they're equal and why.

Q25 Combining what you learned in this section, write a small program that, given any two shape-compatible matrices A and B, plots each rank-one component $C_k = a_k @ b(k)$ as a heatmap subplot, arranged in the style of Figure 10.

Q20 Write a function `random_rank_one_matrix(m, n)` that generates a random $m \times n$ rank-one matrix as the outer product of two random vectors, and use NumPy to verify its rank.

Q22 For the $4 \times 3 @ 3 \times 5$ example in Q7, count how many rank-one matrices were superposed, and confirm this number equals the canceled dimension p .

Q24 Implement `matrix_multiplication_inner` (previous section) and `matrix_multiplication_outer` (this section) side by side, and verify on at least 10 random A, B pairs that their results match exactly.

10 Prompts to Explore Further (1/2)

1

Explain why matrix multiplication $A @ B$ can be written as a sum of p outer-product matrices, and how this connects to and differs from the inner-product expansion view of the previous section.

2

Prove that every outer product $a @ b^T$ is a rank-one matrix, and explain what “rank 1” means geometrically.

3

Explain why every column of $a @ b^T$ is a multiple of a and every row is a multiple of b^T , deriving this from the definition of matrix multiplication.

4

Explain why $a @ b^T$ is always a zero matrix when a or b is the zero vector, and why the matrix's rank is 0 in that case rather than 1.

5

Explain the relationship among rank, linear dependence, and linear independence, and use geometric intuition to explain why the rows/columns of a rank-one matrix must be linearly dependent.

10 Prompts to Explore Further (2/2)

6

Explain why, in the heatmap visualization, superposing the 12 rank-one matrices $a_k @ b(k)$ produces exactly $A @ B$, and how this relates to the outer-product expansion formula.

7

Explain why this section's outer-product implementation (`matrix_multiplication_outer`) needs only a single for-loop, while the previous section's inner-product view needed two nested loops.

8

Explain the dual relationship between the first view of matrix multiplication (inner-product expansion) and the second view (outer-product expansion) — what different aspects of matrix multiplication does each one reveal?

9

Give examples of how rank-one matrix decomposition is used in practice, such as in data compression and low-rank approximation (e.g., principal component analysis).

10

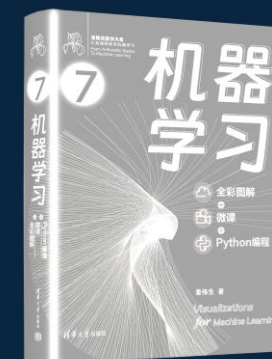
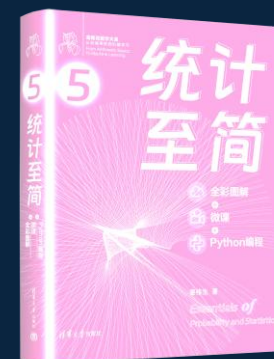
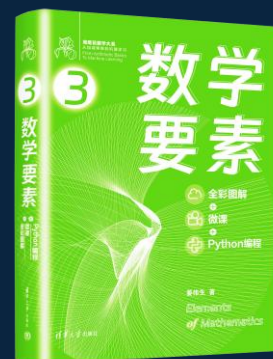
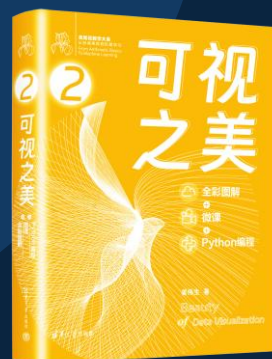
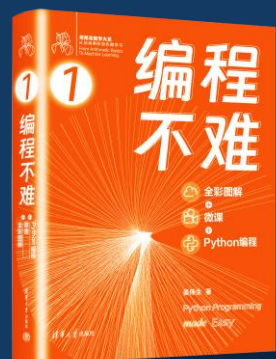
Explain why understanding “matrix multiplication as a sum of rank-one matrices” is essential for later topics such as matrix rank, linear dependence, and matrix decomposition.

数学

数学基础

线性代数

概率统计



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工具

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图和网络

回归、降维
分类、聚类

实践

